# Fit Passport

**One body. One fit identity. Any store.**

*A consumer-owned fit layer for apparel. Project prospectus — 2026.*


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> **This document is a student-project prospectus, not an offer of securities.** It describes the problem, the product as built today, the people we build it for, and where it goes next. Figures describing usage are targets and illustrations, not reported results.

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## 1. The problem

**Sizes do not agree, and never have.** A medium in one brand is a large in another; a 32 waist measures anywhere from 31 to 34 inches of actual cloth; a jacket that fits your shoulders swims at the chest. The label is a guess dressed up as a fact.

The cost of that guess lands on two parties:

- **The shopper** wastes money, time, and trust. Online apparel returns run around a quarter of everything sold, and *"wrong size / wrong fit"* is the single largest reason. Every return is a shipping cost, a refund cycle, and a small erosion of confidence that makes the next purchase harder.
- **Everyone downstream** pays for it — the retailer eats reverse logistics, and the returned garment is often landfilled rather than resold.

The information needed to solve this **already exists** — it is scattered across the one place nobody has organised it: *the clothes that already fit you.* Your closet is a dataset of ground truth about your body, brand by brand, and today it is thrown away at the moment of every new purchase.

**Fit Passport turns that closet into a portable profile you own** — and translates it to any product page you paste.

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## 2. The product

Fit Passport is a web application. There is nothing to install. It has five parts, and the first four all exist and work today.

### 2.1 The Passport

Your fit identity as a single object: a metal charge-card — portrait, holder, region, preferred fit, a membership number (`No. 00000001`), and a verification line. Its **finish is themed by your highest**

**earned badge**, and you can set it to any metal you've earned. It exports as a PNG and reappears as your banner across the community, so a colour reads as *a person* everywhere in the app. Precise measurements never appear on it.

### 2.2 The Closet

Your ground-truth wardrobe. Add items by pasting a product URL — the extractor reads brand, garment type, gender, and size chart from the whole link, not just the last path segment. Organise into colour-coded collections in a **filing-cabinet view**: stacked file cards you pull out onto a "desk," a comparison bucket to set items aside, edit history, and honest per-garment fit ratings. This is the dataset the engine learns you from.

### 2.3 The Size Check

Paste any product link — bare domains are fine. The engine reads the size chart and **ranks every available size, with a per-signal reason for each.** A live multi-region converter sits alongside it. Crucially:

> **The engine is a transparent rule-and-score model, not a black box and not an LLM.** Every recommendation shows its reasoning. An LLM is used only to *extract* product data from a page — it never decides your size. This is a deliberate, load- bearing choice: a fit recommendation you can't interrogate is a fit recommendation you can't trust, and trust is the entire product.

### 2.4 The Community

The reason to come back. An opt-in directory of members (each wearing their metal banner), an outfit feed with likes, a **follow graph and a followed feed**, and **Ask & Answer** — questions anchored in *real closets*, where an answer can attach a garment you actually own as evidence. Leaderboards surface the day's top looks and the most genuinely helpful members. Everything social is opt-in; everything is moderated (report, block, an admin review queue, rate limits).

### 2.5 The Badges

The prestige layer that makes contribution worth it — an earned-only ladder of struck-metal medals (see the companion *Badge System* design document). Status here is scarce and cannot be bought, which is precisely what makes people log honest data and help strangers.

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## 3. Why this is defensible

A size calculator alone is a feature, not a company — it can be cloned in a weekend. The moat is not the calculator. It is what the calculator *creates*:

1. **Verified fit context.** Every member has a real closet with brands, sizes, and honest fit ratings. So "this looks good" becomes **"this fits a body like mine, in this size, from this brand."** No general fashion feed can say that, because no general fashion feed knows what actually fits its users.

1. **Earned-only prestige.** Badges and the metal card are derived from real behaviour. Status is visible, scarce, and non-purchasable — a durable reason to contribute that a competitor can't buy their way past.

1. **Crowd fit-knowledge that improves the engine.** "This brand runs small on broad shoulders" is knowledge the community generates and the engine absorbs. Every honest rating and answered question makes the core recommendation better — a compounding, data-driven advantage that widens with scale. **This is the real moat.**

1. **The privacy posture is a feature, not a cost.** Your account code lets someone read your closet and a *coarse* body type — **never your precise measurements.** That constraint is enforced in the data model itself, not by policy, and it's what lets fit be *social* without being invasive.

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## 4. Who we build for

Three overlapping audiences, in priority order:

| AUDIENCE | THE JOB THEY HIRE US FOR | WHY THEY STAY |
|---|---|---|
| **The uncertain buyer** | "Will this actually fit me before I pay?" | Fewer returns, more confidence — the entry drug |
| **The taste-builder** | "I want to dress better and learn how" | A feed of buyable, wearable looks with provenance |
| **The taste-sharer / blogger** | "I want my eye to be seen and to matter" | Earned status; a following built on being right, not loud |

The wedge is the **uncertain buyer** — the pain is sharp, frequent, and universal. The *retention* comes from turning that buyer, over time, into a taste-builder and then a taste-sharer. The badge ladder is the rail that carries them along that path.

We deliberately do **not** show scraped brand imagery or logos — brand names appear as text only, and the only photos on the platform are ones users uploaded themselves. This is a considered trademark-and-copyright position, and it doubles as a trust signal: this is *your* data, not a repackaging of the brands'.

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## 5. What is built today

Fit Passport is a working application, not a mockup. As of this writing:

| AREA | STATUS |
| --- | --- |
| Transparent fit engine with per-signal reasoning | **Built** |
| URL extractor (brand / garment / gender / size chart) | **Built** |
| Closet with collections, filing-cabinet view, edit history | **Built** |
| Passport metal card, badge-themed, PNG export, member numbers | **Built** |
| Live multi-region size converter | **Built** |
| Outfits, likes, follow graph + followed feed | **Built** |
| Ask & Answer with real-closet evidence | **Built** |
| Leaderboards (daily top looks, top members) | **Built** |
| Badge system — 20 medals, dimensional, turn-to-inspect, WebGL | **Built** |
| Anonymous session → claim → shareable account code | **Built** |
| Moderation: report, block list, admin review queue, rate limits | **Built** |
| Coverage: automated tests across engine, badges, extractor, converters | **115 passing** |

**Architecture.** Next.js (App Router) · React · TypeScript · Tailwind · Prisma · Zod · Vitest, with Framer Motion + Lenis for motion and a lazily-loaded three.js for the badge inspect view only. Local development runs on SQLite with **no API keys and no cloud services**; production runs on Vercel + Neon Postgres, with the Postgres schema *derived* from the same source so the two can't drift. Sessions are HMAC-signed cookies verified byte-identically on both the Node and Edge runtimes.

**Design system.** Black-led, on a cool porcelain workspace, with a single cobalt accent spent sparingly, and a characterful editorial serif (Fraunces) paired with a clean sans (Inter) — both self-hosted so a deploy never depends on a font CDN. The aesthetic target is *fashion-forward, high-end, simple, usable* — taste as a first-class requirement, because our users have taste and expect the tools they use to have it too.

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## 6. How it could make money

Not yet monetised — this is a course-stage MVP — but the model is legible and the data asset points at several honest options, in rough order of alignment:

- **Affiliate on the buy.** When we tell you the right size and you buy with confidence, an affiliate link is a natural, non-intrusive cut that aligns us with *the purchase working out* rather than with ad impressions.
- **A fit API for retailers.** The same engine, offered to a store as "will this fit this shopper?", directly attacks their single largest returns cost. Our per-brand fit-truth is the pitch.

- **Premium membership.** Deeper analytics, unlimited closet history, advanced badges and card finishes — paid for by the taste-builders and taste-sharers who get the most from the ecosystem.

Every path is downstream of the same asset: **structured, honest, consumer-owned fit data at scale.** We build that first; monetisation follows the data.

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## 7. Where it goes next

Sequenced smallest-first, so each step proves demand before the next is built:

1. **Deepen the community loop** — richer Ask & Answer, the followed feed as the default home, daily top looks. *(Largely built; now about density.)*
2. **Run one budget styling contest manually** — a $100 "Thrift Run" — to test whether the fashion audience shows up *before* building event tooling.
3. **Budget contests as a system** — $100 weekly, $1,000 monthly, $10,000 seasonal; itemised-price looks, community voting, commemorative special-metal badges for winners. A budget makes taste comparable and levels the field.
4. **Fit-matched discovery** — filter any feed by "people shaped like me," using the coarse public body type, never measurements. Needs member density to matter.
5. **The retailer fit API** — once the crowd fit-knowledge is deep enough to sell.

The guardrails never move: everything social stays opt-in, precise measurements never become social or matching data, and no step relaxes the privacy invariant no matter how useful it would be.

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## 8. Team & context

Built by **Xiangchen Kong** and **Alyssa Qi** for Carnegie Mellon's *49-800 Start Up Creation in Practice* (Fall 2026), advised by **Prof. Sheryl Root**. The project is a functioning MVP and a live demo, developed under a detailed engineering log, with a Business Model Canvas, Value Proposition Canvas, customer-interview guide, and a risk/legal review maintained alongside the code.

> The bet, in one line: **the closet is the dataset, the passport is the interface, and the community is the moat.**

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*Companion documents: the Badge System design document (this folder); the community ecosystem plan, deployment runbook, and identity/threat model (`docs/`); the engineering log (`DEVLOG.md`); and the source itself (`app-web/`).*